
CS246: Database Management Systems Lab

Lab # 13 (1 Questions, 55 Points)

Held on 07-Apr-2025

Lab Timings: 14:00 to 17:00 Hours Pages: 3

Submission: 17:00 Hrs, 07-Apr-2025

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- a. This lab assignment is based on the concepts covered in chapter 5 **Advanced SQL** in the CS245 theory class.
- b. You can refer to the text book for SQL syntax.

Question 1: (55 points)

Pivot tables and OLAP functions Perform the following tasks using MySQL :

Task 01 (1 mark) Create a database named *week13*

Task 02 (3 marks) **Create tables**

- a. A **location** table containing the following

1 st column	location_id	integer
2 nd column	city	fixed length char of size 10
3 rd column	state	fixed length char of size 2
4 th column	country	fixed length char of size 20

with `location_id` as primary key.

- b. A **product** table containing the following

1 st column	product_id	integer
2 nd column	product_name	fixed length char of size 10
3 rd column	category	fixed length char of size 2
4 th column	price	integer

with `product_id` as primary key.

- c. A **sale** table containing the following

1 st column	product_id	integer
2 nd column	time_id	integer
3 rd column	location_id	integer
4 th column	sales	integer

with `product_id`, `time_id`, `location_id` as primary key.

Task 03 (3 marks) **populate data**

- a. Populate data from file `location.csv` into table `location`
- b. Populate data from file `product.csv` into table `product`
- c. Populate data from file `sale.csv` into table `sale`

Task 04 (48 marks) Building a **pivot table**

	WI	CA	Total
1995	63	81	144
1996	38	107	145
1997	75	35	110
Total	176	223	399

year_state_01

a. (12 marks) Construct a pivot table `year_state_01` - *method - 01*

- Whose columns are **states** WI, CA, total
- Whose rows are **years** 1995, 1996, 1997, total

by writing the following *individual queries* to construct the pivot table

- Compute the total sales for the state WI in the year 1995 - Cell(1,1)
- Compute the total sales for the state CA in the year 1995 - Cell(1,2)
- Compute the total sales in the year 1995 for the states (WI, CA) - Cell(1,3)
- Compute the total sales for the state WI in the year 1996 - Cell (2, 1)
- Compute the total sales for the state CA in the year 1996 - Cell (2, 2)
- Compute the total sales in the year 1996 for the states (WI, CA) - Cell (2, 3)
- Compute the total sales for the state WI in the year 1997 - Cell (3, 1)
- Compute the total sales for the state CA in the year 1997 - Cell (3, 2)
- Compute the total sales in the year 1997 for the states (WI, CA) - Cell (3, 3)
- Compute the total sales for the states WI in the years (1995, 1996, 1997) - Cell (4, 1)
- Compute the total sales for the states CA in the years (1995, 1996, 1997) - Cell (4, 2)
- Compute the total sales for the states (WI, CA) in the years (1995, 1996, 1997) - Cell(4, 3)

xiii. The result of all the above queries should be a `year_state` pivot table

b. (12 marks) Construct a pivot table `year_state_02` - *method - 02*

	WI	CA	Total
1995	63	81	144
1996	38	107	145
1997	75	35	110
Total	176	223	399

year_state_02_01 year_state_02_02

year_state_02_03 year_state_02_04

- i. (3 marks) Write a single query using `sale`, `location` tables to generate `year_state_02_01`
- ii. (3 marks) Write a single query using `year_state_02_01` table to generate `year_state_02_02`
- iii. (3 marks) Write a single query using `year_state_02_01` table to generate `year_state_02_03`
- iv. (3 marks) Write a single query using either `year_state_02_02` or `year_state_02_03` to generate `year_state_02_04`
- c. (12 marks) Construct a pivot table `year_state_03` same as pivot table `year_state_01`
- *method - 03*
 - Compute the pivot table through a **single query**.
 - Hint 1: The query would involve `group by` over year
 - Hint 2: For each column of the pivot table, use `case` statement and `sum` aggregation function
- d. (12 marks) Construct a pivot table `year_state_04` same as pivot table `year_state_01`
- *method - 04*
 - Compute the pivot table through `rollup` operation